

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	04 July 2026
Team ID	
Project Name	Plugging into the Future: An Exploration of Electricity Consumption Patterns
Maximum Marks	4 Marks

Functional Requirements :

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement	Description
FR-1	Data Import	Import the electricity consumption dataset into Tableau Desktop.
FR-2	Data Preprocessing	Clean, validate, and prepare the dataset for analysis.
FR-3	Data Visualization	Create charts and visualizations to analyze electricity consumption.
FR-4	Dashboard Development	Develop interactive dashboards using Tableau Desktop.
FR-5	Interactive Filters	Enable users to filter data by Year, State, Region, Quarter, Month, Lockdown Period, and Recovery Phase.
FR-6	Tableau Story	Create a Tableau Story to present the analysis in a logical sequence.
FR-7	Dashboard Publishing	Publish the dashboard on Tableau Public for online access.

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The dashboard should be easy to use and understand with interactive visualizations.
NFR-2	Security	The dataset should be stored securely and protected from unauthorized modifications.
NFR-3	Reliability	The dashboard should consistently display accurate and error-free analysis results.

NFR No.	Non-Functional Requirement	Description
NFR-4	Performance	The system should load datasets quickly and generate visualizations with minimal delay.
NFR-5	Availability	The Tableau dashboard should be accessible whenever users need to analyze electricity consumption data.
NFR-6	Scalability	The system should support larger datasets, additional years, and more states or regions without significantly affecting dashboard performance.